

LOSO (glove_ml_explorer) Experiment SB_flex_ypr

Static · flex + YPR (both hands), no Nothing
 Generated 2026-05-12 02:06:48

Experiment configuration

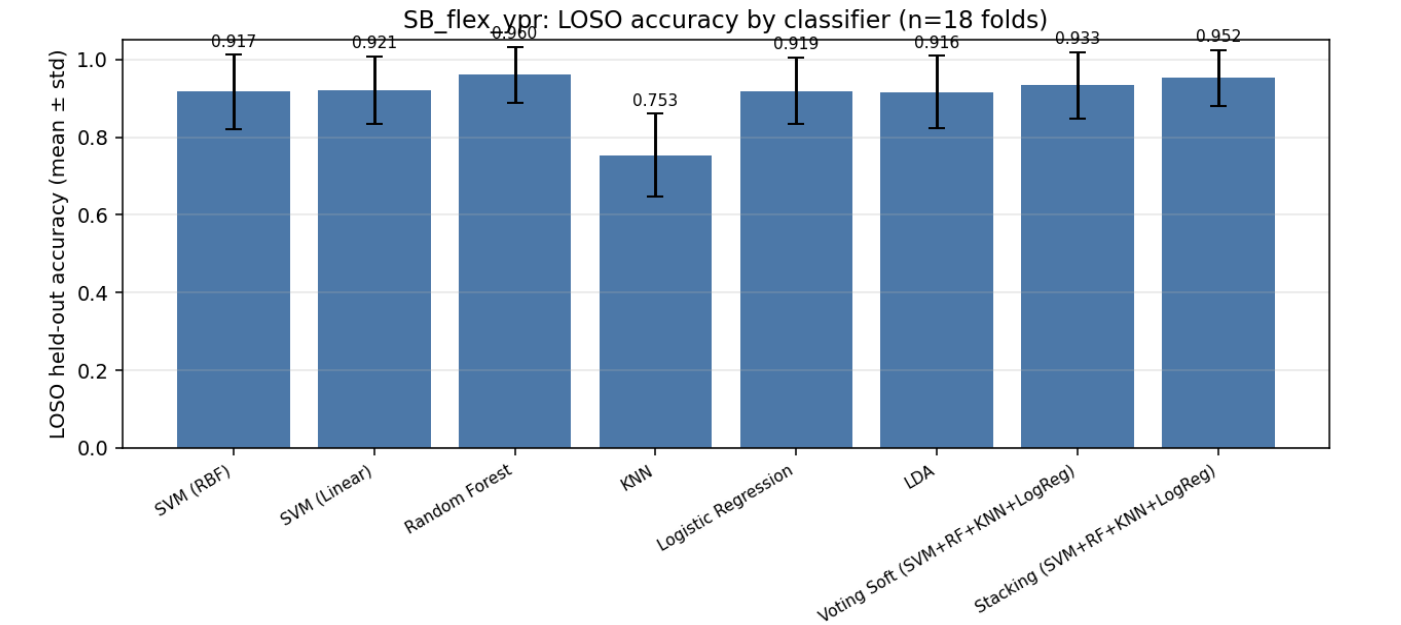
Class set	6 classes excluding Double_Nothing
Classes	Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman
Sensor modalities	flex, YPR
Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Sensor channels	86
Sequence length	90 steps (resampled)
Filter	off
Feature mode	stats+fft
Normalisation	minmax
Augmentation	off
Inner CV folds	5
LOSO subjects	18: 10_Josh, 11_Mansh, 12_Marcus, 13_Nat, 14_StephenV2, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 5_Georgia, 6_Harry, 7_Henry, 8_Jestin, 9_Joselyn, Stephen, Val

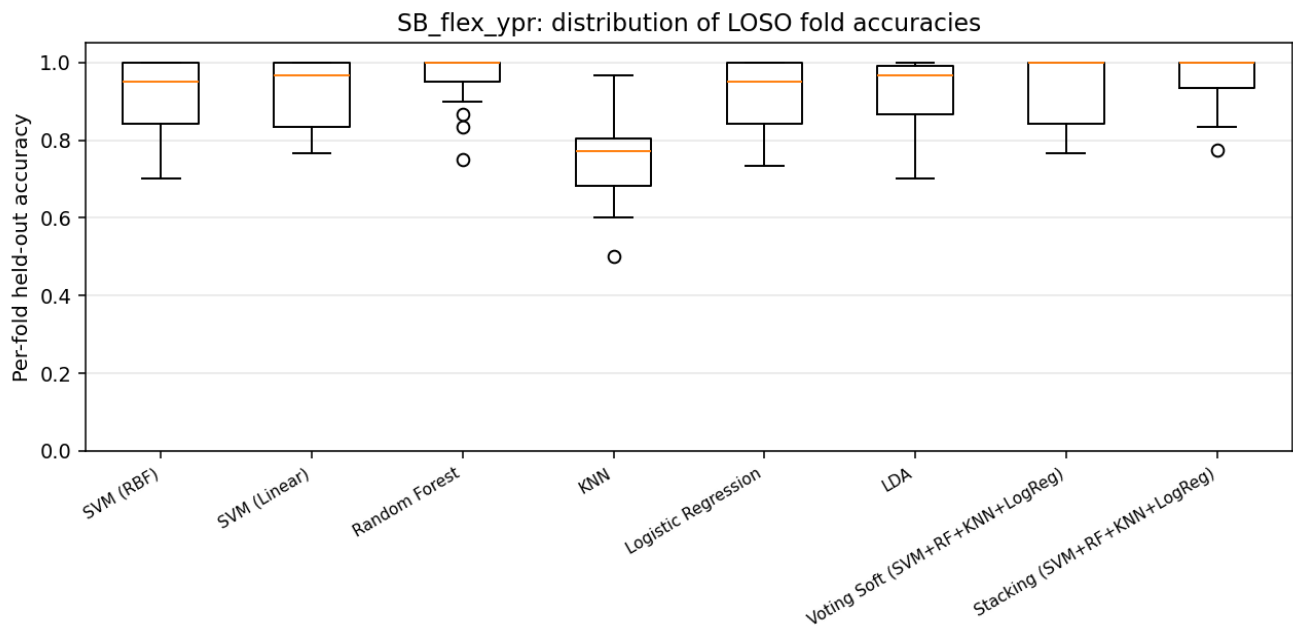
LOSO summary across all subjects

Classifier	Params*	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean
SVM (RBF)	—	0.9171	0.0957	0.7000	1.0000	0.9799
SVM (Linear)	69,675	0.9211	0.0876	0.7667	1.0000	0.9795
Random Forest	14,998	0.9602	0.0723	0.7500	1.0000	0.9766
KNN	—	0.7530	0.1071	0.5000	0.9667	0.8724
Logistic Regression	27,870	0.9189	0.0862	0.7333	1.0000	0.9761
LDA	27,870	0.9163	0.0931	0.7000	1.0000	0.9485
Voting Soft (SVM+RF+KNN+LogReg)	—	0.9334	0.0861	0.7667	1.0000	0.9809
Stacking (SVM+RF+KNN+LogReg)	—	0.9523	0.0719	0.7750	1.0000	0.9835

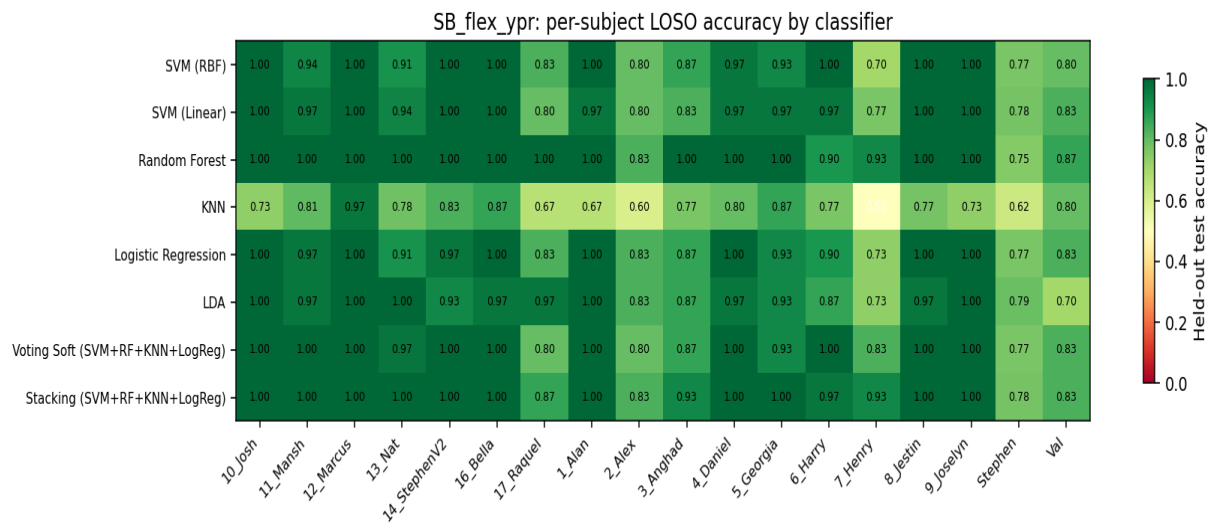
*Params: linear models use coef_+intercept size; tree ensembles use total node count across trees. SVM-RBF / KNN / Voting / Stacking are non-parametric or composite, so they show as —.

Best classifier by LOSO mean: **Random Forest**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Stacking (SVM+RF+KNN+LogReg)
10_Josh	604	1.000	1.000	1.000	0.733	1.000	1.000	1.000	1.000
11_Mansh	603	0.935	0.968	1.000	0.806	0.968	0.968	1.000	1.000
12_Marcus	604	1.000	1.000	1.000	0.967	1.000	1.000	1.000	1.000
13_Nat	602	0.906	0.938	1.000	0.781	0.906	1.000	0.969	1.000
14_StephenV2	604	1.000	1.000	1.000	0.833	0.967	0.933	1.000	1.000
16_Bella	604	1.000	1.000	1.000	0.867	1.000	0.967	1.000	1.000
17_Raquel	604	0.833	0.800	1.000	0.667	0.833	0.967	0.800	0.867
1_Alan	604	1.000	0.967	1.000	0.667	1.000	1.000	1.000	1.000
2_Alex	604	0.800	0.800	0.833	0.600	0.833	0.833	0.800	0.833
3_Anghad	604	0.867	0.833	1.000	0.767	0.867	0.867	0.867	0.933
4_Daniel	604	0.967	0.967	1.000	0.800	1.000	0.967	1.000	1.000
5_Georgia	604	0.933	0.967	1.000	0.867	0.933	0.933	0.933	1.000
6_Harry	604	1.000	0.967	0.900	0.767	0.900	0.867	1.000	0.967
7_Henry	604	0.700	0.767	0.933	0.500	0.733	0.733	0.833	0.933
8_Jestin	603	1.000	1.000	1.000	0.774	1.000	0.968	1.000	1.000
9_Joselyn	604	1.000	1.000	1.000	0.733	1.000	1.000	1.000	1.000
Stephen	514	0.767	0.775	0.750	0.625	0.767	0.792	0.767	0.775
Val	604	0.800	0.833	0.867	0.800	0.833	0.700	0.833	0.833

Classifier: SVM (RBF)

Params (best-effort): —

LOSO mean accuracy: 0.9171 ± 0.0957 (min 0.7000, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	4/6	5/5	5/5	5/5	5/5	5/5	0.935
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	4/5	3/5	6/6	5/5	0.906
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	0/5	5/5	5/5	5/5	5/5	0.833
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	4/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	4/5	5/5	2/5	5/5	5/5	5/5	0.867
4_Daniel	5/5	5/5	5/5	4/5	5/5	5/5	0.967
5_Georgia	5/5	3/5	5/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
7_Henry	5/5	2/5	1/5	5/5	3/5	5/5	0.700
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	6/20	18/20	13/20	18/20	19/20	18/20	0.767
Val	1/5	5/5	4/5	4/5	5/5	5/5	0.800
MEAN	0.887	0.872	0.881	0.894	0.975	0.994	0.917

Classifier: SVM (Linear)

Params (best-effort): 69,675

LOSO mean accuracy: 0.9211 ± 0.0876 (min 0.7667, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	5/6	5/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	3/5	6/6	5/5	0.938
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	0/5	5/5	4/5	5/5	5/5	0.800
1_Alan	5/5	4/5	5/5	5/5	5/5	5/5	0.967
2_Alex	5/5	4/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	4/5	5/5	2/5	4/5	5/5	5/5	0.833
4_Daniel	4/5	5/5	5/5	5/5	5/5	5/5	0.967
5_Georgia	5/5	5/5	4/5	5/5	5/5	5/5	0.967
6_Harry	5/5	5/5	5/5	5/5	4/5	5/5	0.967
7_Henry	5/5	4/5	1/5	5/5	3/5	5/5	0.767
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	5/20	19/20	14/20	18/20	19/20	18/20	0.775
Val	2/5	5/5	4/5	4/5	5/5	5/5	0.833
MEAN	0.894	0.908	0.883	0.883	0.964	0.994	0.921

Classifier: Random Forest

Params (best-effort): 14,998

LOSO mean accuracy: 0.9602 ± 0.0723 (min 0.7500, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	5/5	5/5	5/5	5/5	5/5	5/5	1.000
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	2/5	5/5	5/5	0.900
7_Henry	5/5	5/5	5/5	5/5	3/5	5/5	0.933
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	4/20	18/20	20/20	12/20	18/20	18/20	0.750
Val	3/5	5/5	5/5	3/5	5/5	5/5	0.867
MEAN	0.933	0.994	1.000	0.867	0.972	0.994	0.960

Classifier: KNN

Params (best-effort): —

LOSO mean accuracy: 0.7530 ± 0.1071 (min 0.5000, max 0.9667)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	0/5	4/5	4/5	5/5	5/5	0.733
11_Mansh	3/6	3/5	5/5	4/5	5/5	5/5	0.806
12_Marcus	4/5	5/5	5/5	5/5	5/5	5/5	0.967
13_Nat	6/6	5/5	3/5	0/5	6/6	5/5	0.781
14_StephenV2	4/5	5/5	3/5	3/5	5/5	5/5	0.833
16_Bella	2/5	5/5	5/5	4/5	5/5	5/5	0.867
17_Raquel	5/5	0/5	5/5	0/5	5/5	5/5	0.667
1_Alan	5/5	0/5	5/5	0/5	5/5	5/5	0.667
2_Alex	5/5	2/5	0/5	1/5	5/5	5/5	0.600
3_Anghad	5/5	4/5	0/5	4/5	5/5	5/5	0.767
4_Daniel	4/5	5/5	4/5	1/5	5/5	5/5	0.800
5_Georgia	5/5	3/5	5/5	3/5	5/5	5/5	0.867
6_Harry	5/5	5/5	1/5	2/5	5/5	5/5	0.767
7_Henry	0/5	5/5	0/5	0/5	5/5	5/5	0.500
8_Jestin	6/6	5/5	3/5	0/5	5/5	5/5	0.774
9_Joselyn	4/5	5/5	2/5	1/5	5/5	5/5	0.733
Stephen	4/20	16/20	7/20	11/20	19/20	18/20	0.625
Val	3/5	1/5	5/5	5/5	5/5	5/5	0.800
MEAN	0.761	0.689	0.631	0.442	0.997	0.994	0.753

Classifier: Logistic Regression

Params (best-effort): 27,870

LOSO mean accuracy: 0.9189 ± 0.0862 (min 0.7333, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	5/6	5/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	3/5	4/5	6/6	5/5	0.906
14_StephenV2	4/5	5/5	5/5	5/5	5/5	5/5	0.967
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	0/5	5/5	5/5	5/5	5/5	0.833
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	4/5	5/5	1/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	2/5	5/5	5/5	5/5	0.867
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	3/5	5/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	5/5	3/5	4/5	5/5	0.900
7_Henry	5/5	3/5	4/5	5/5	0/5	5/5	0.733
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	6/20	18/20	14/20	18/20	18/20	18/20	0.767
Val	1/5	5/5	5/5	4/5	5/5	5/5	0.833
MEAN	0.885	0.883	0.917	0.906	0.928	0.994	0.919

Classifier: LDA

Params (best-effort): 27,870

LOSO mean accuracy: 0.9163 ± 0.0931 (min 0.7000, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	5/6	5/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	4/5	5/5	4/5	5/5	5/5	5/5	0.933
16_Bella	5/5	5/5	5/5	4/5	5/5	5/5	0.967
17_Raquel	5/5	5/5	5/5	4/5	5/5	5/5	0.967
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	1/5	4/5	5/5	0.833
3_Anghad	3/5	5/5	5/5	3/5	5/5	5/5	0.867
4_Daniel	5/5	5/5	5/5	4/5	5/5	5/5	0.967
5_Georgia	5/5	4/5	4/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	4/5	2/5	5/5	5/5	0.867
7_Henry	5/5	4/5	2/5	4/5	2/5	5/5	0.733
8_Jestin	6/6	4/5	5/5	5/5	5/5	5/5	0.968
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	3/20	18/20	19/20	19/20	18/20	18/20	0.792
Val	2/5	4/5	5/5	1/5	4/5	5/5	0.700
MEAN	0.877	0.950	0.931	0.808	0.939	0.994	0.916

Classifier: Voting Soft (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9334 ± 0.0861 (min 0.7667, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	4/5	6/6	5/5	0.969
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	0/5	5/5	4/5	5/5	5/5	0.800
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	4/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	4/5	5/5	2/5	5/5	5/5	5/5	0.867
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	3/5	5/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
7_Henry	5/5	5/5	3/5	5/5	2/5	5/5	0.833
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	5/20	19/20	14/20	18/20	18/20	18/20	0.767
Val	1/5	5/5	5/5	4/5	5/5	5/5	0.833
MEAN	0.903	0.908	0.928	0.906	0.961	0.994	0.933

Classifier: Stacking (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9523 ± 0.0719 (min 0.7750, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	1/5	5/5	5/5	5/5	5/5	0.867
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	4/5	5/5	5/5	5/5	0.933
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	4/5	5/5	5/5	0.967
7_Henry	5/5	5/5	5/5	5/5	3/5	5/5	0.933
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	6/20	19/20	14/20	18/20	18/20	18/20	0.775
Val	1/5	5/5	5/5	4/5	5/5	5/5	0.833
MEAN	0.906	0.953	0.972	0.917	0.972	0.994	0.952